



LABORATORY MANUAL

SC4172
IoT - TinyML

Lab Experiment #2

Deployment of Model

SESSION 2025/2026
SEMESTER 2

COLLEGE OF COMPUTING AND DATA SCIENCE
NANYANG TECHNOLOGICAL UNIVERSITY

1. **OBJECTIVES**

Continuation from Lab #1, where the trained model of the cosine predictor is to be deployed on the Arduino Nano 33 BLE Sense board.

2. **LABORATORY**

This experiment is conducted at the **Software Laboratory 1 (Location: N4-01a-02)**.

3. **HARDWARE EQUIPMENT**

- Computer installed with Anaconda and Tools for TensorFlow ML.
- Arduino Nano 33 BLE Sense board
- Arduino IDE installed with TensorFlow Lite (TFLite) library

4. **REFERENCES**

- TinyML book by Pete Warden, Daniel Situnayake. 2019.
<https://www.oreilly.com/library/view/tinyml/9781492052036/>
- Nano33 BLE Sense Board. <https://docs.arduino.cc/hardware/nano-33-ble-sense/>
- SC4172 Lecture Notes

5. **INTRODUCTION**

After the TensorFlow model has been trained to have satisfactory performance, the model will need to be further processed in order to enable it to be deployed on the microcontroller board.

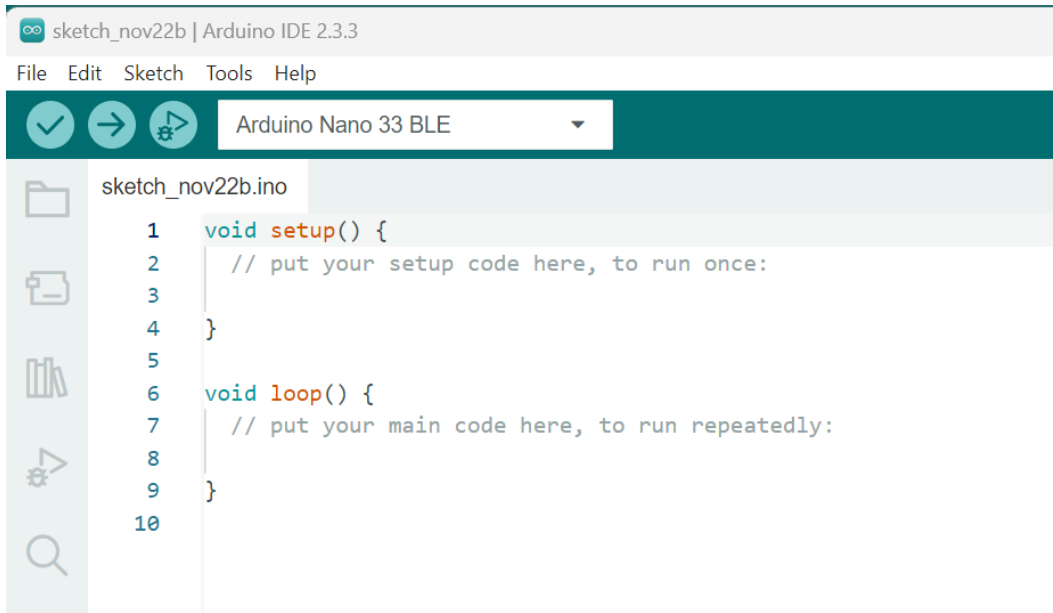
There are two further steps required to achieve this

- Convert the trained model to the TFLite model and stored as TFLite for Microcontroller format.
- Integrate the model with the application software and deploy it on the nano 33 Sensor board.

6. ARDUINO SETUP

Arduino IDE is available in two major versions: IDE1.x and IDE2.x. Both versions should work with the application used in the SC4172 Labs. The illustration in this manual correspond to IDE2.x. An Arduino program consists of the following fixed structure:

- `setup()` routine that initializes the system with the necessary code.
- `loop()` routine that execute the program logic in an endless loop



i. General Setup

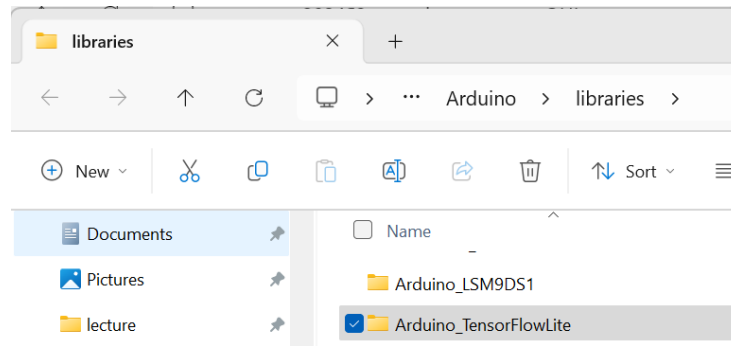
The hardware board is selected via [Tools->Board](#) menu. Search for “[Arduino Mbed OS Nano Boards](#)”.

The [COM Port](#) connecting to the board is selected via [Tools->Port](#) menu. This is done after selecting the hardware board, which is [Arduino Nano 33 BLE](#) in this case.

You can add [libraries](#) via [Tools->Managed Libraries...](#) menu. You can also manually add libraries by adding the library code directly to the Arduino Library Folder. More on this in the next sub section.

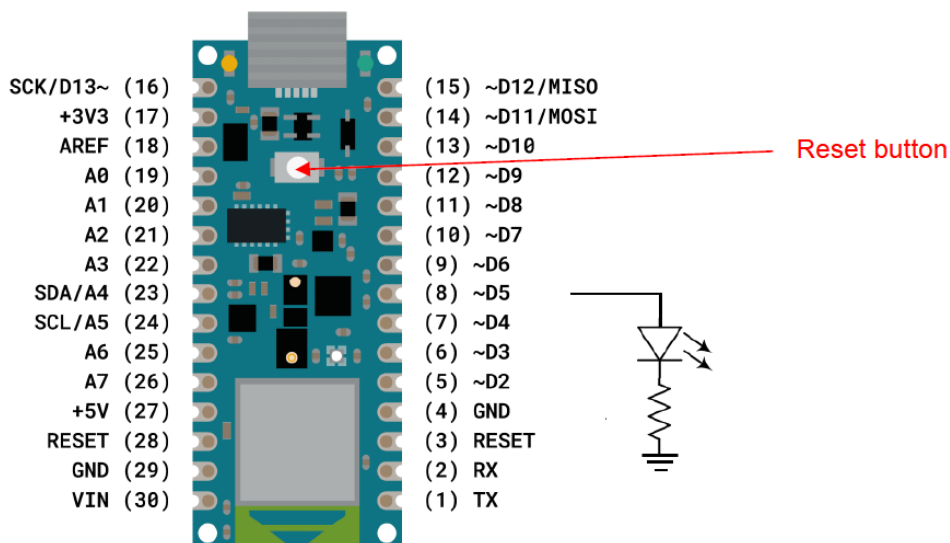
ii. Arduino Tensorflow Lite Library Setup

Arduino Tensorflow Lite library is not available in the Library Manger so needs to be added manually. Download the Tensorflow Lite library (**Arduino_TensorFlowLite-2.4.0-ALPHA**) from NTU Learn, unzip, rename folder to “[Arduino_TensorFlowLite](#)” and copy the folder to the Documents\Arudino\Libraries Folder in the Windows/Mac. You will be able to see Tensorflow Lite related examples in Arduino IDE’s examples.



7. NANO 33 SENSE BOARD

The Arduino Nano 33 BLE Sense combines a tiny form factor, different environment sensors and the possibility to run AI using TinyML and TensorFlow™ Lite. Below is a pinout diagram of the Nano 33 Sense Board.



- i. Run the Blinky.ino application from NTULearn pin the Arduino IDE to test the interface between your PC and the nano 33 sense board. The red and blue LED should blink alternatively if interface is ok.
- ii. Note the following information
 - The compilation of code in Arduino IDE is slow, especially slow during the first compilation. This is likely due to the anti-virus software interfering and slowing down the compilation operation.
 - Explore and check which pin corresponds to which colour LED. There are a few LEDs on board. Check the header file `arduino_pins.h` for details.

8. PORTING TENSORFLOW LITE TO NANO 33

With the C header file created in your Google CoLab or Anaconda, open the Arduino Sketch “[Hello_World_Sine_Skeleton](#)” downloaded from NTU Learn.

Fill in the missing code to allow the Arduino Sketch to run inferences based on the model provided in the C Header file. **You will need to refer to the lecture notes for the detail operations involved.**

If the code is integrated correctly, you should see `y_val` (prediction of $\cos(x_val)$), based on some `x_val` input. Output depends on whether the ‘[DEBUG](#)’ label is one or zero. If `DEBUG=0`, one of the LED, depending on which pin you choose to use (remember to initial [led_pin](#)), should pulsate at a fix frequency, simulating `y_val` when `x_val` goes cycle through 0 to 2π radians. Some text output can be seen via the [Serial Monitor](#) as well. See if you can also enable a cosine wave to be seen on the [Serial Plotter](#).

HINT:

- `x_val` input is between 0 and 2π radians
- `y_val` output is between 0 and 1
- You can use the Arduino function `analogWrite(led_pin, brightness)` lit up the LED with different brightness. ‘brightness’ input to the function is a value between 0 and 255 and related to the PWM duty cycle. Check online for more details.

9. UNDERSTANDING THE CODE

Go through the code again to understand the role of Tensorflow Lite related functions and header files.